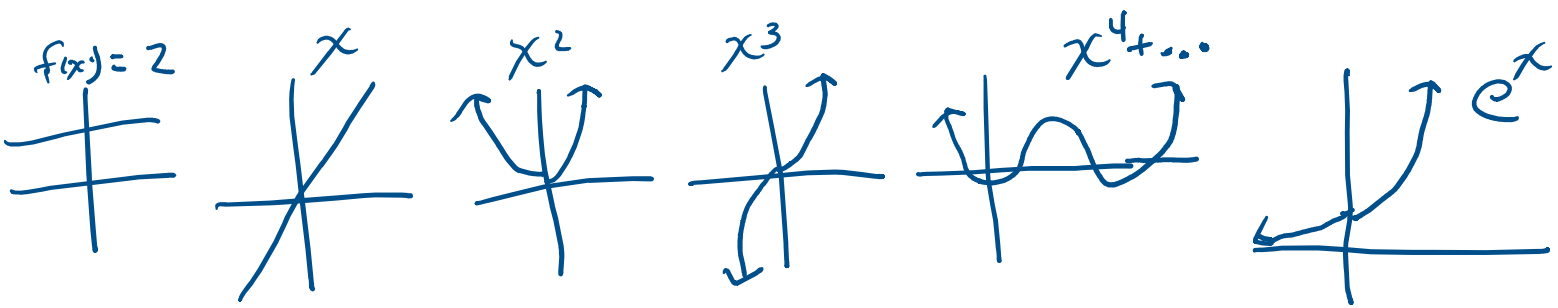
Rational functions and indirect variation

Up until now, all the functions we have been looking at are continuous functions: constant, linear, quadratic, polynomial, and exponential.
A continuous function can be graphed without taking your pencil off the paper.
Now we will be talking about functions where there is a variable in the denominator or a rational expression. Some of these functions are continuous, but many are not.



Rational functions have restrictions on the domain: x cannot be any number that makes the denominator zero. $\frac{1}{x}$ $x \neq 0$

A rational function has variables in the denominator. Because the denominator cannot equal zero, the variable can not equal a number that makes the denominator zero. We can explicitly write $x \neq 0$, for y=1/x but…

Rather than constantly writing out restrictions on the domain (what x cannot equal), all restrictions on the domain are implied for this PPT unit.

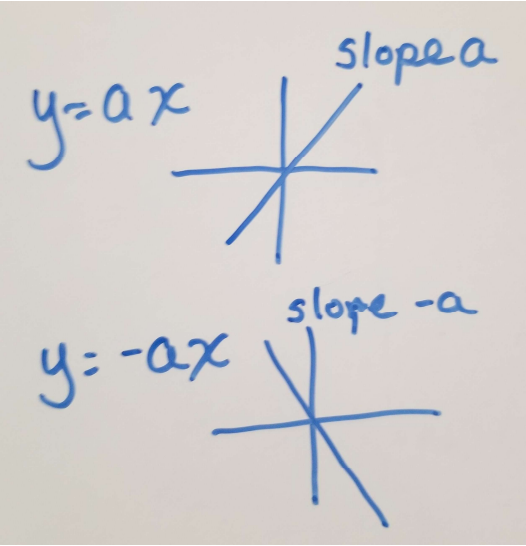
I will not explicitly write that the variable cannot equal a number that makes the denominator equal zero but know that the restriction is there!

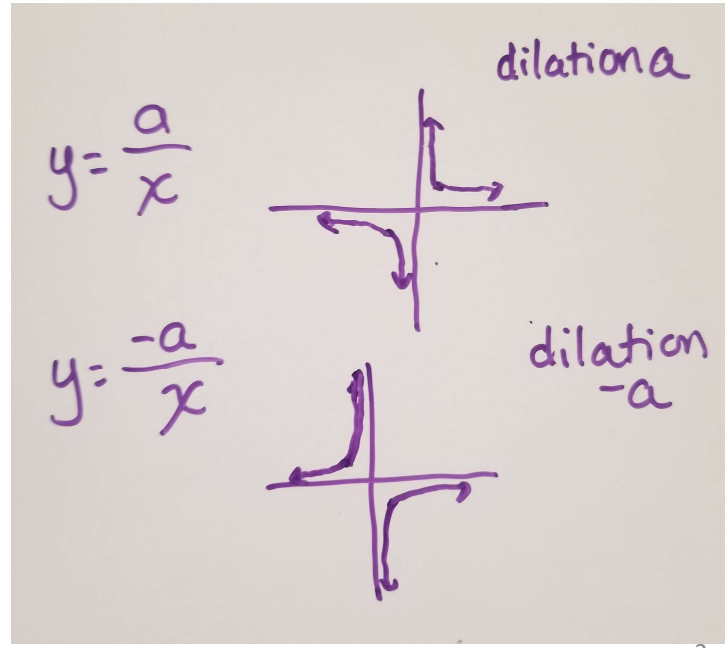
Think of this as a consent notice or a waiver that the denominator cannot equal zero.

Direct variation is when the output is a constant multiple of the input; when the output is proportional to the input.

Indirect variation is when the input times the output gives a constant, so the output varies with the reciprocal of the input. $x \cdot y = \text{constant}$

The output is inversely proportional to the input.

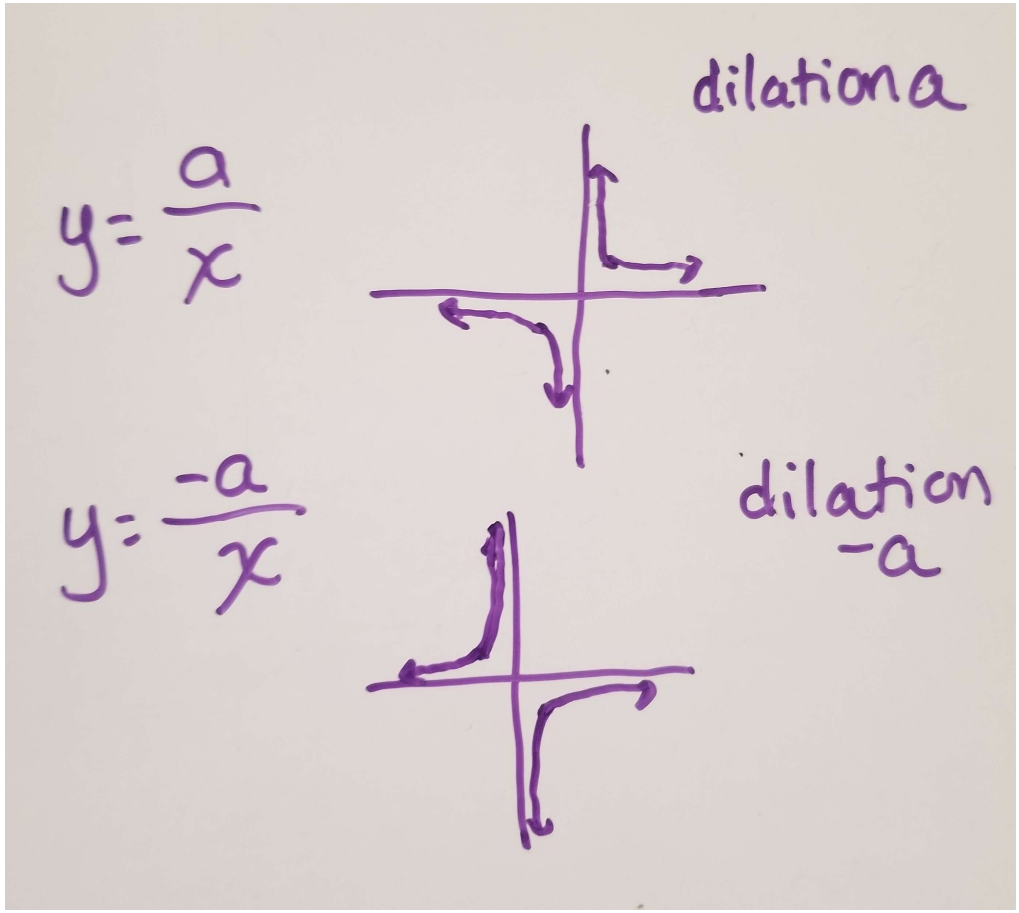




Indirect variation is when the output gets bigger as the input gets smaller and vice versa.

$$a = yx$$

$$y = \frac{a}{x} \quad x \neq 0$$

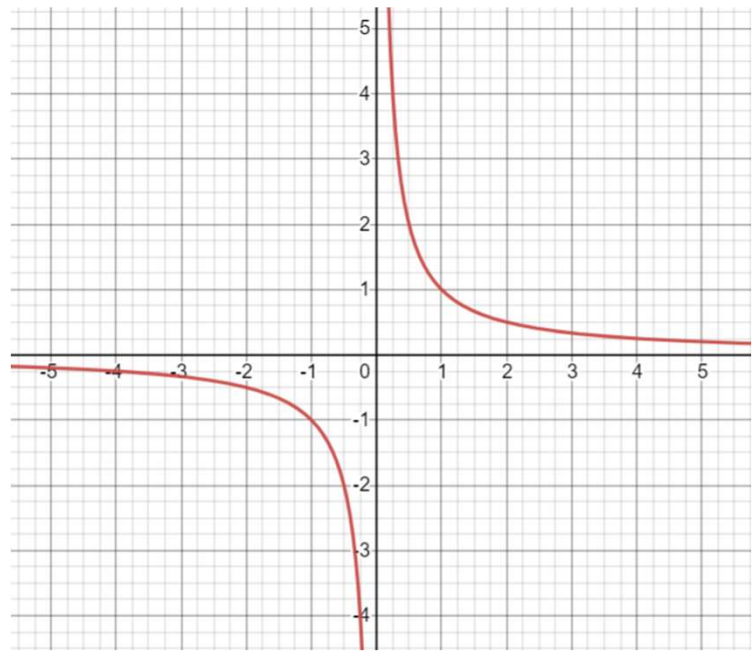


$$y = \frac{1}{x}$$

A rational function has a variable in the denominator. Since there can never be a zero in the denominator, this can create some strange behavior. Rational functions can be different from each other, so we don't have a parent function to model them.

We can start by looking at the rational function f(x) = $\frac{1}{x}$ or y= $\frac{1}{x}$

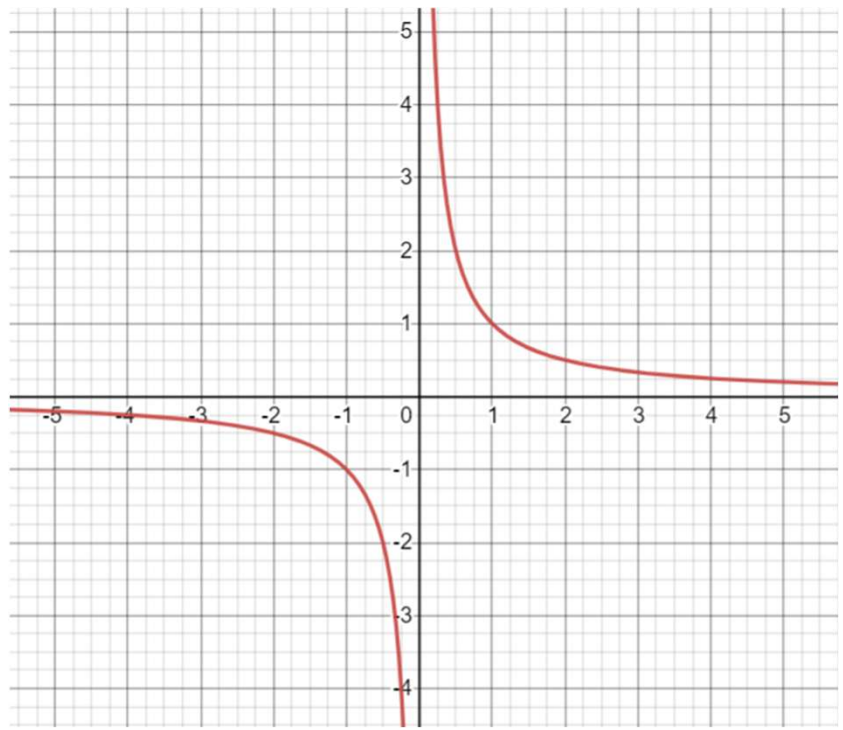
$$y = \frac{1}{x}$$



f(x)=$\frac{1}{x}$ or y= $\frac{1}{x}$

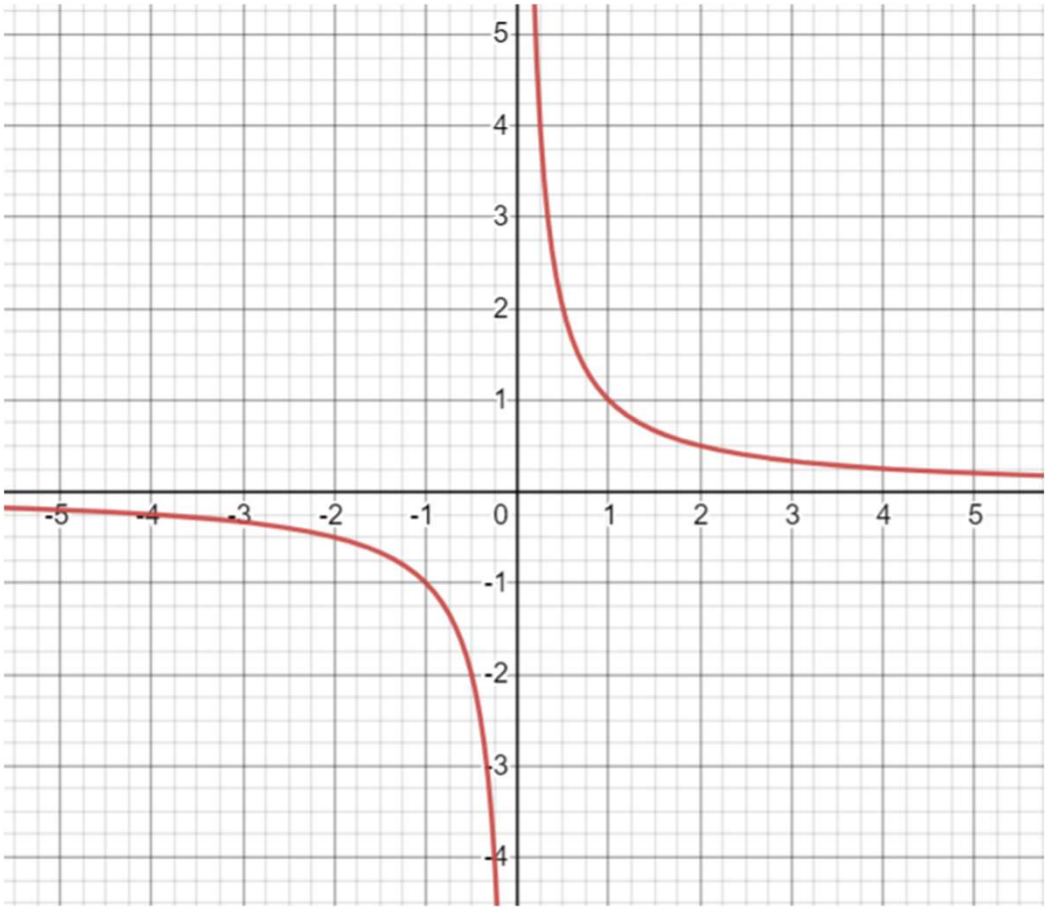
Like our parent linear function y=x, the output has the same sign as the input. This is an odd function as it reflects about the origin.

$y = \frac{1}{x}$



When there is a variable in the denominator, there can't be input that makes the denominator zero. So, restrictions are placed on the domain.

f(x)= $\frac{1}{x}$ or y= $\frac{1}{x}$ $x \neq 0$

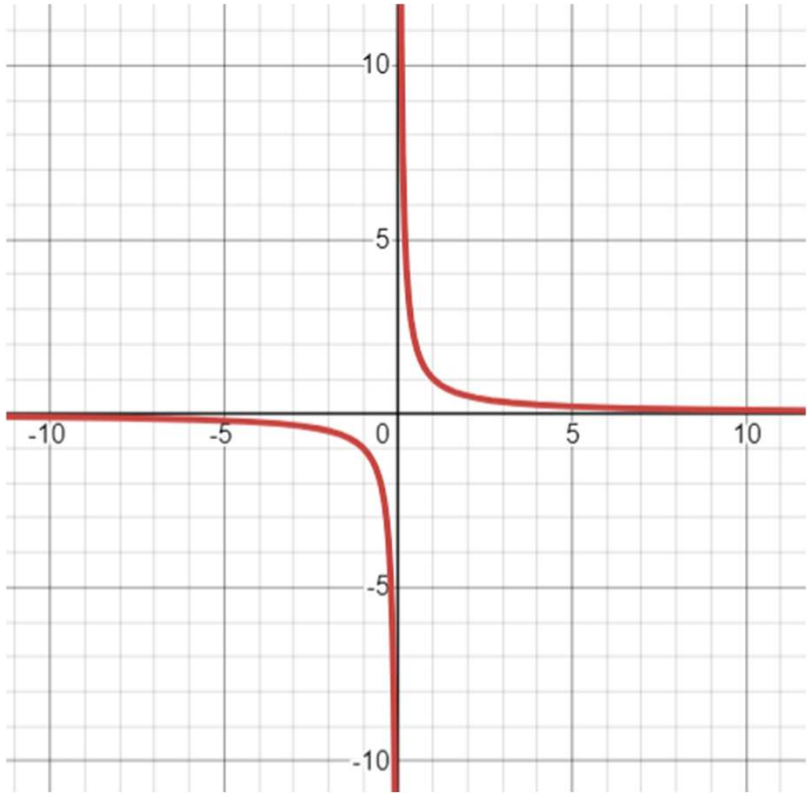


$y = \frac{1}{x}$

When there is a variable in the denominator, there can't be input that makes the denominator zero. In this example, you can see horizontal and vertical **asymptotes**.

$y = \frac{1}{x}$

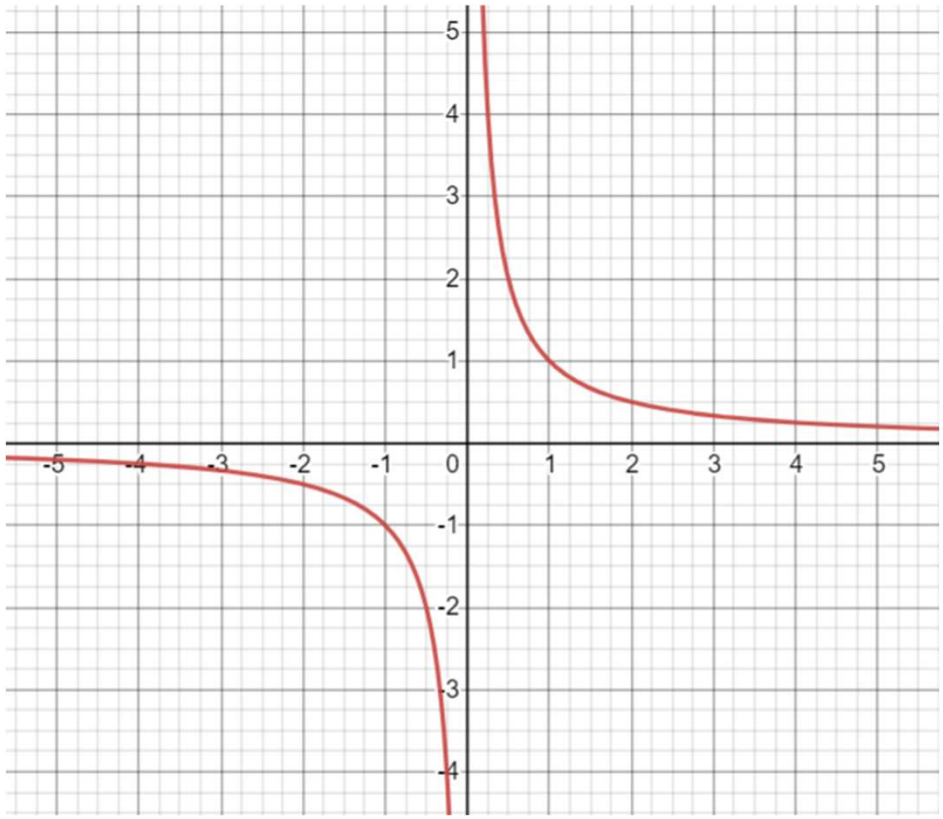
f(x)=$\frac{1}{x}$ or y= $\frac{1}{x}$ x$\neq 0$



Vertical asymptotes occur when the output goes up or down to infinity and there is a discontinuity of the function. Horizontal asymptotes occur when the input goes to infinity and the output goes towards a line.

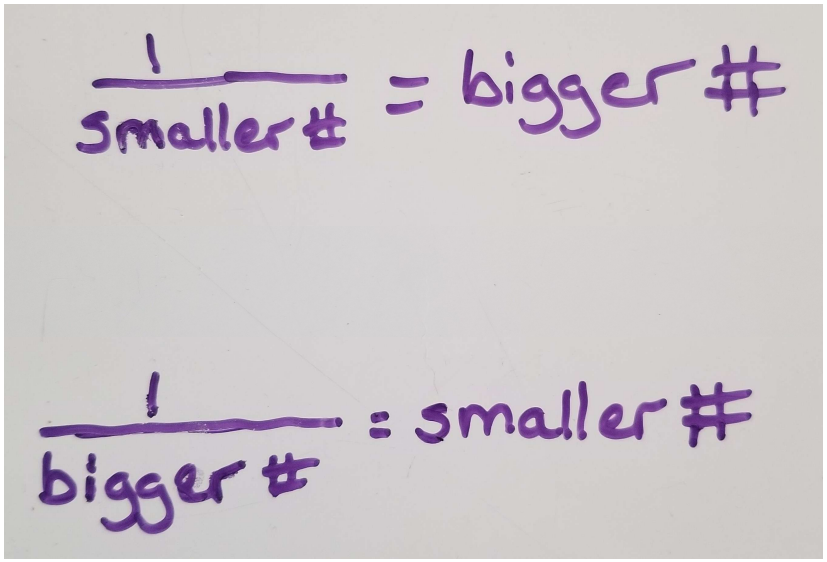
$y = \frac{1}{x}$

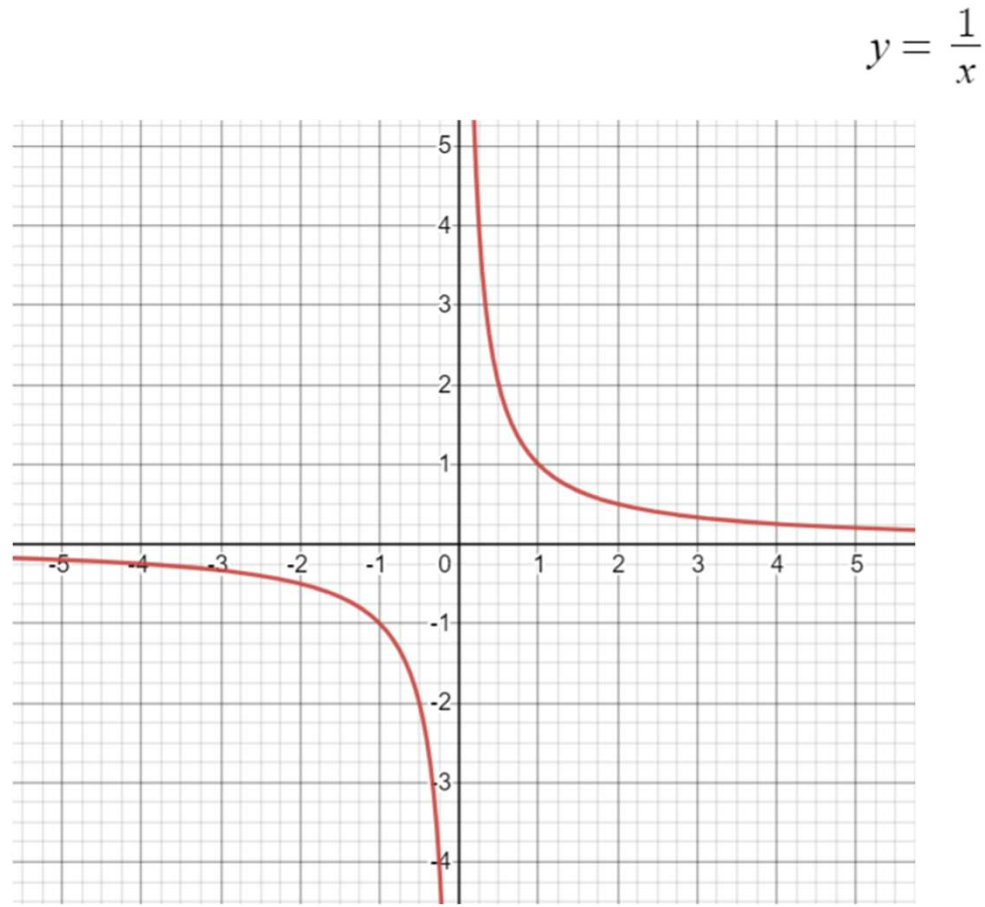
f(x)=$\frac{1}{x}$ or y= $\frac{1}{x}$ x$\neq 0$



A hyperbola is a type of conic section from geometry, like an ellipse or a parabola. Conic section – Wikipedia

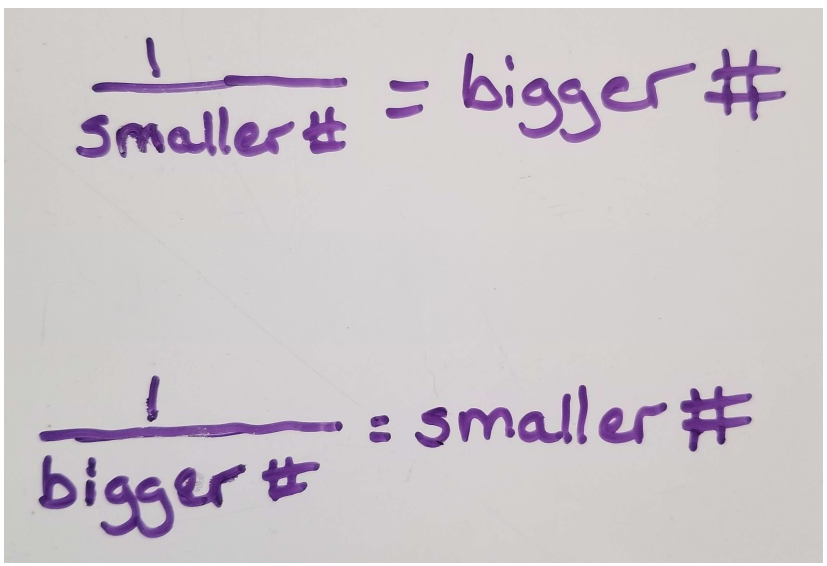
f(x)=$\frac{1}{x}$ or y= $\frac{1}{x}$ x≠0

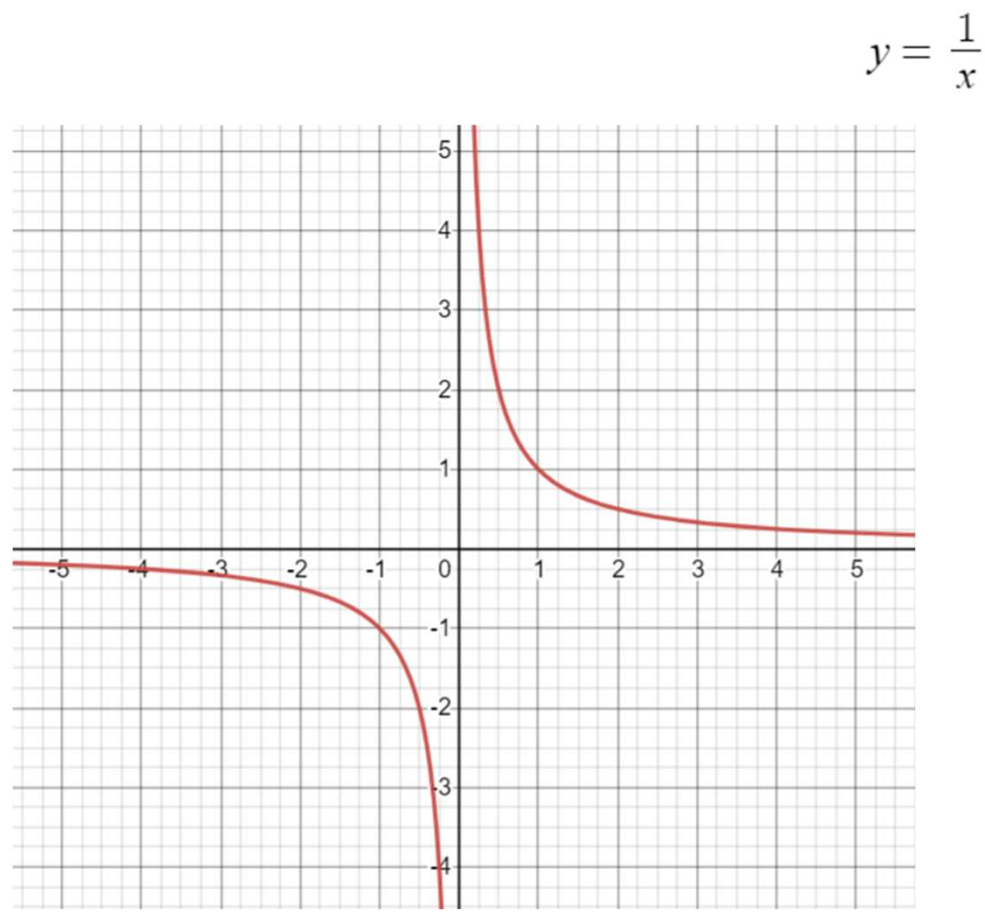




For y=1/x, this pattern is a **hyperbola**. A vertical asymptote occurs when x is zero or at the y axis. A horizontal asymptote occurs when y is zero or at the x axis.

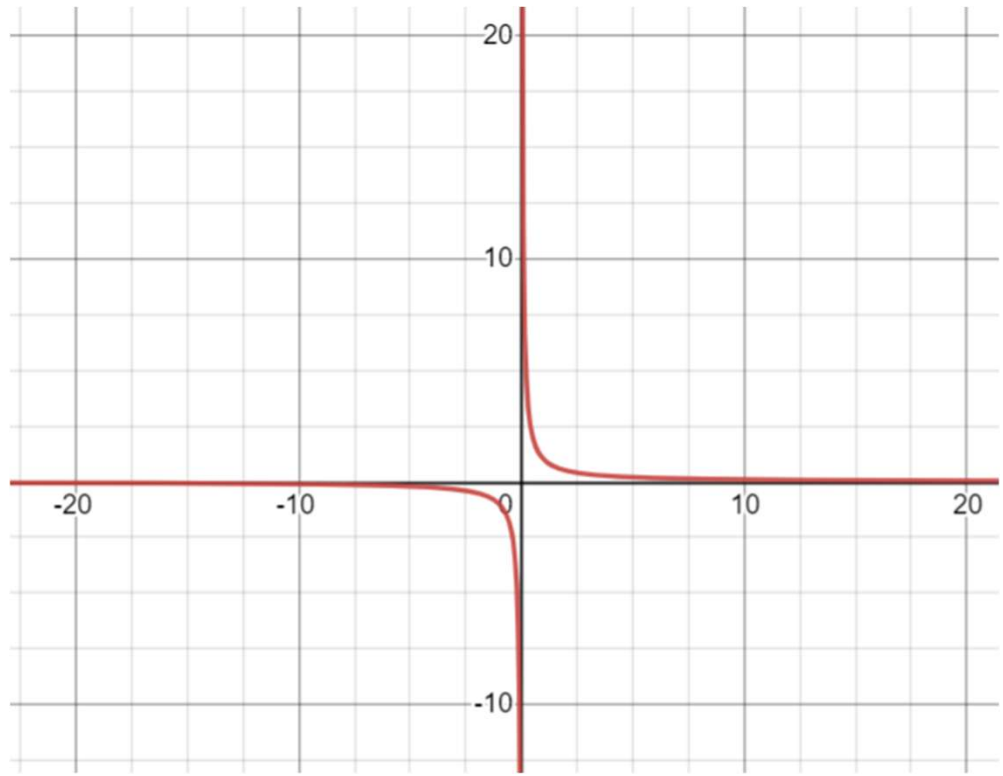
f(x)=$\frac{1}{x}$ or y= $\frac{1}{x}$ $x \neq 0$





Vertical asymptotes occur when the output goes up or down to infinity and there is a discontinuity of the function. Horizontal asymptotes occur when the input goes to infinity and the output goes towards but never reaches a horizontal line.

f(x)=$\frac{1}{x}$ or y= $\frac{1}{x}$ x$\neq$0



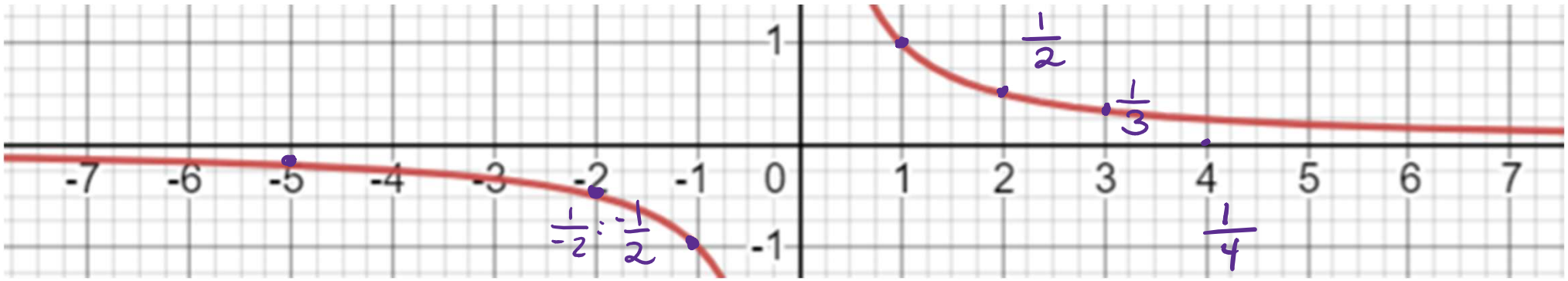
$y = \frac{1}{x}$

Horizontal asymptotes occur when the input goes to infinity and the output goes towards a line.

$f(x)=\frac{1}{x}$ or $y=\frac{1}{x}$ $x\neq 0$

$\frac{1}{\text{bigger \#}}$ = smaller #

$y = \frac{1}{x}$



$\frac{1}{-5} = -\frac{1}{5}$ $\frac{1}{-1} = -1$ $\frac{1}{1} = 1$ $\frac{1}{\text{million}}$ is tiny $\frac{1}{10^6}$

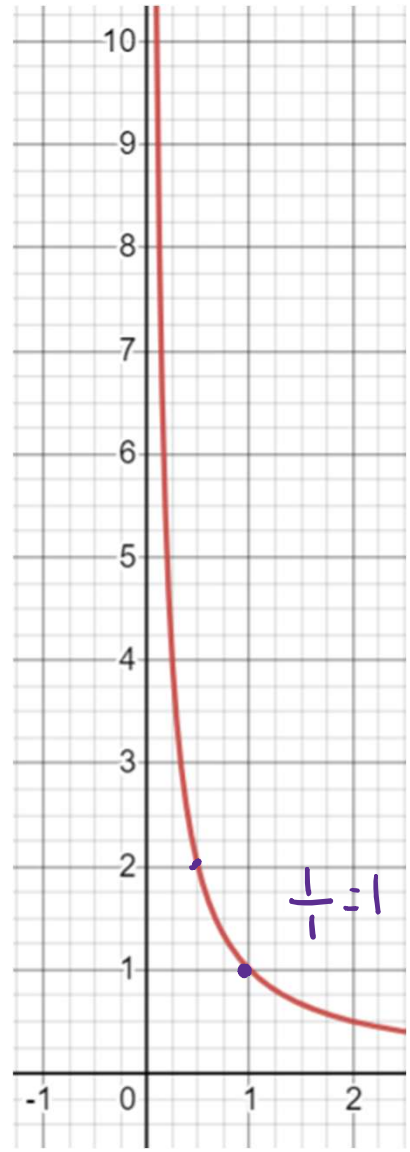
$\frac{1}{-\infty}$ = gets infinitely tiny

Focusing in on when x is positive and getting smaller and smaller.

As x gets smaller from one to zero, the output gets bigger and bigger.

f(x)=$\frac{1}{x}$ or y= $\frac{1}{x}$ $x \neq 0$

$$\frac{1}{\text{smaller \#}} = \text{bigger \#}$$



$y = \frac{1}{x}$

$\frac{1}{\frac{1}{2}} = 2$

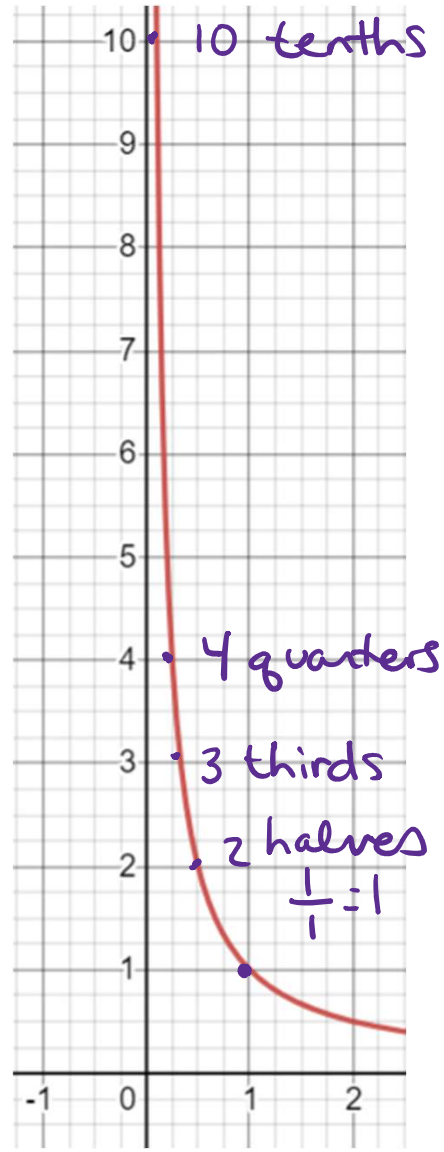
$\frac{1}{\frac{1}{4}} = 4$

Dividing by fractions:

$$\frac{1}{\text{smaller \#}} = \text{bigger \#}$$

$$\frac{1}{\frac{1}{2}} = \text{how many halves?}$$

halves



$$y = \frac{1}{x}$$

$$\frac{1}{\frac{1}{2}} = 2$$

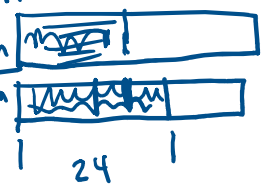
$$\frac{1}{\frac{1}{4}} = 4$$

It can get tricky to visualize a fraction divided by a fraction.

$$\frac{\frac{1}{2}\ \text{yd}}{\frac{2}{3}\ \text{yd}}$$

$\frac{18\text{ in}}{36\text{ in}}$ 18

$\frac{24\text{ in}}{36}$



$\frac{1\text{ wh}}{2\text{ pcs}}$

$\frac{2\text{ pcs wh}}{3\text{ pcs}}$

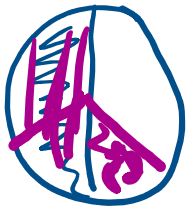
$\frac{18}{24} = \frac{3}{4}$ of the $\frac{2}{3}$ yd gives $\frac{1}{2}$ yd

24 in 18 in

$$\frac{3}{4} \cdot \cancel{\frac{2}{3}} = \frac{1}{2} \cdot \frac{3}{2}$$

$$\frac{\cancel{3}}{\cancel{4}} \cdot \frac{\cancel{2}}{\cancel{3}} = \frac{1}{2} \checkmark$$

$$\frac{3}{4} \cdot \cancel{\frac{2}{3}} = \frac{\frac{1}{2}}{\frac{2}{3}}$$



how much of the $\frac{2}{3}$ is $\frac{1}{2}$

$\frac{3}{4}$ $\frac{1}{2}$ is $\frac{3}{4}$ of the $\frac{2}{3}$

Since division is the inverse of multiplication, dividing by a fraction is the same as multiplying by the reciprocal. Reciprocal is the special term for the multiplicative inverse of a number..

$$\frac{\frac{1}{2}}{\frac{2}{3}} \qquad \frac{1}{2} \cdot \frac{3}{2} = \frac{3}{4} \qquad \frac{\frac{a}{b}}{\frac{c}{d}} = \frac{a}{b} \cdot \frac{d}{c} \qquad \frac{2}{3} \sim \frac{3}{2} \qquad \frac{2}{3} \cdot \frac{3}{2} = 1$$

$$\frac{a \ \frac{\text{mult}}{\text{div}}}{c \ \frac{\text{div}}{\text{mult}}} \qquad \frac{\frac{\text{mult } a}{\text{div } b}}{\frac{\text{div } c}{\text{mult } d}} \qquad \frac{a \cdot d}{b \cdot c} \ \begin{matrix}\text{mult} \\ \text{div}\end{matrix}$$

Since division is the inverse of multiplication, dividing by a fraction is the same as multiplying by the reciprocal. Reciprocal is the special term for the inverse of a fraction. $\frac{2}{3} \curvearrowright \frac{3}{2}$ $\frac{2}{3} \cdot \frac{3}{2} = 1$

$$\frac{\frac{1}{2}}{\frac{2}{3}} \qquad \frac{1}{2} \cdot \frac{3}{2} = \frac{3}{4}$$

$$\frac{\left(\frac{1}{2}\right)^{-1}}{\left(\frac{2}{3}\right)^{-1}} = \frac{\left(1^{+1} \cdot 2^{-1}\right)^{+1}}{\left(2^{+1} \cdot 3^{-1}\right)^{-1}} = 1^{+1} \cdot 2^{-1} \cdot 2^{-1} \cdot 3^{+1} = \frac{1 \cdot 3}{2 \cdot 2} = \frac{3}{4}$$

$$\frac{\frac{1}{2} \quad \frac{\text{top}}{\text{bottom}} \quad \frac{\text{hi}}{\text{lo}}}{\frac{2}{3} \quad \frac{\text{bottom}}{\text{top}} \quad \frac{\text{lo}}{\text{hi}}} = \frac{1 \cdot 3}{2 \cdot 2} = \frac{3}{4}$$

When there are multiple operations, it can get complicated!

$$\frac{\frac{a}{b}}{c} = \frac{a}{bc}$$

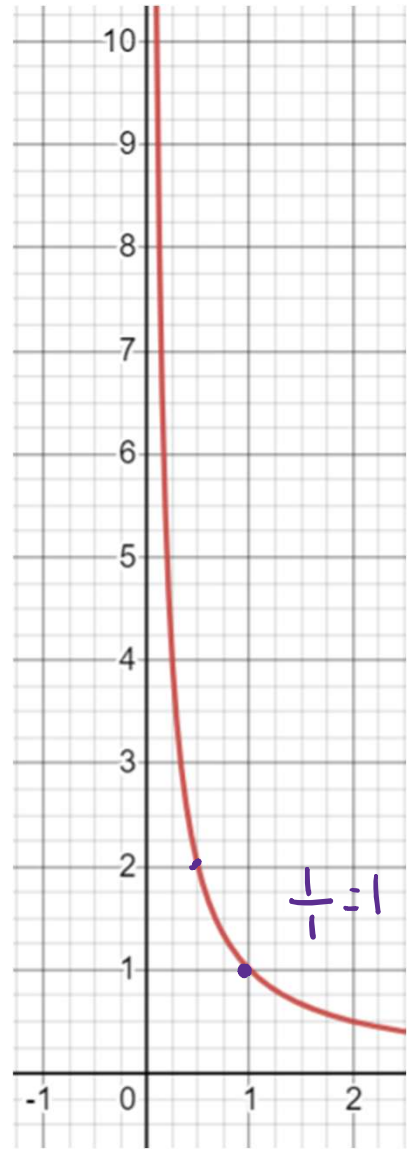
$$\frac{\frac{\frac{a}{b}}{\frac{c}{d}}}{\frac{\frac{m}{n}}{\frac{p}{q}}} \qquad \frac{\frac{ad}{bc}}{\frac{mq}{np}} = \frac{ad\,np}{bc\,mq}$$

Focusing in on when x is positive and getting smaller and smaller.

As x gets smaller from one to zero, the output gets bigger and bigger.

f(x)=$\frac{1}{x}$ or y= $\frac{1}{x}$ $x \neq 0$

$$\frac{1}{\text{smaller \#}} = \text{bigger \#}$$



$y = \frac{1}{x}$

$\frac{1}{\frac{1}{2}} = 2$

$\frac{1}{\frac{1}{4}} = 4$

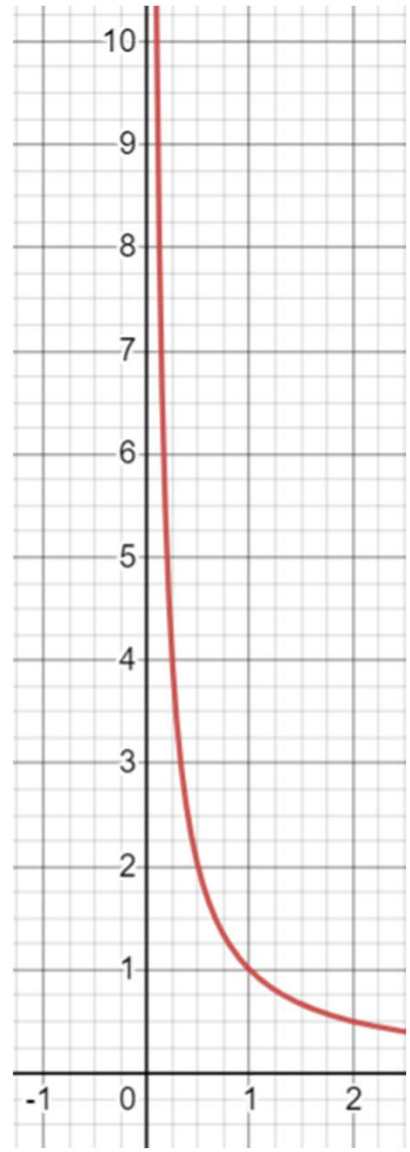
$\frac{1}{\frac{1}{10}} = 10$

Focusing in on when x is positive and getting smaller and smaller.

As x gets smaller from one to zero, the output gets bigger and bigger.

f(x)=$\frac{1}{x}$ or y= $\frac{1}{x}$ $x \neq 0$

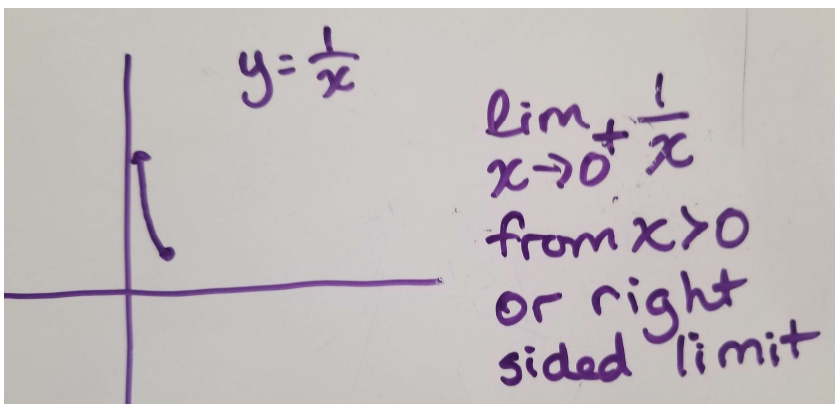
$\frac{1}{\text{tiny}}$ = big

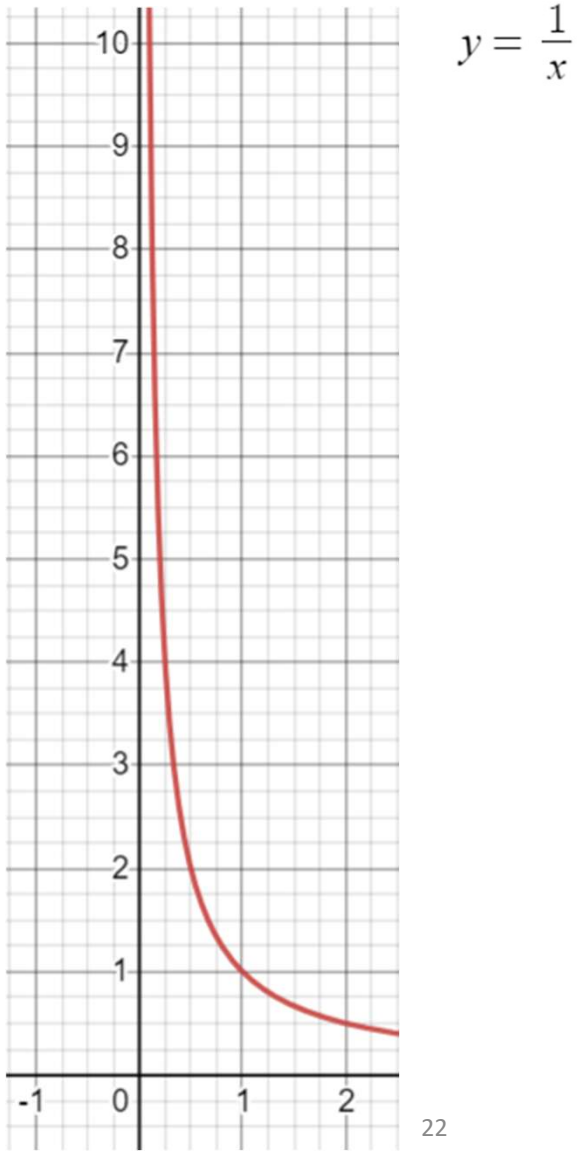


$y = \frac{1}{x}$

1/x diverges at x=0. As x gets smaller from one to zero, the output gets bigger and bigger. The function can never reach x=0 but as it gets closer, coming in from the right to the left, the output gets infinitely big. We say the limit as x approaches zero from the right to the left is positive infinity. This is a one-sided, right-hand limit.

f(x)=$\frac{1}{x}$ or y= $\frac{1}{x}$ $x \neq 0$



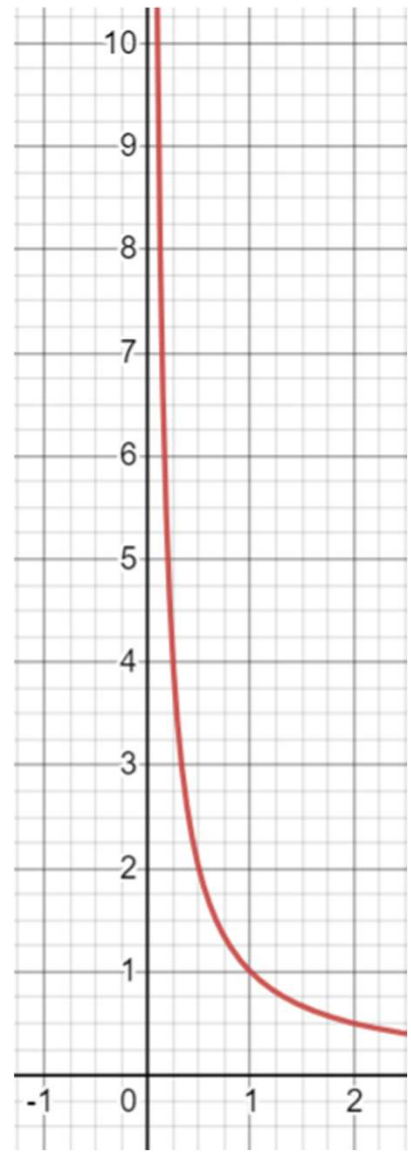


The limit as x approaches zero from the right to the left is positive infinity. For me, it is easier to see and follow from a graph, than looking at it algebraically.

f(x)=$\frac{1}{x}$ or y= $\frac{1}{x}$ x$\neq 0$

$$\lim_{x \to 0^+} \frac{1}{x} = +\infty$$

$$\frac{1}{\text{smaller \#}} = \text{bigger \#}$$



$y = \frac{1}{x}$

Focusing in on when x is negative and getting smaller and smaller.

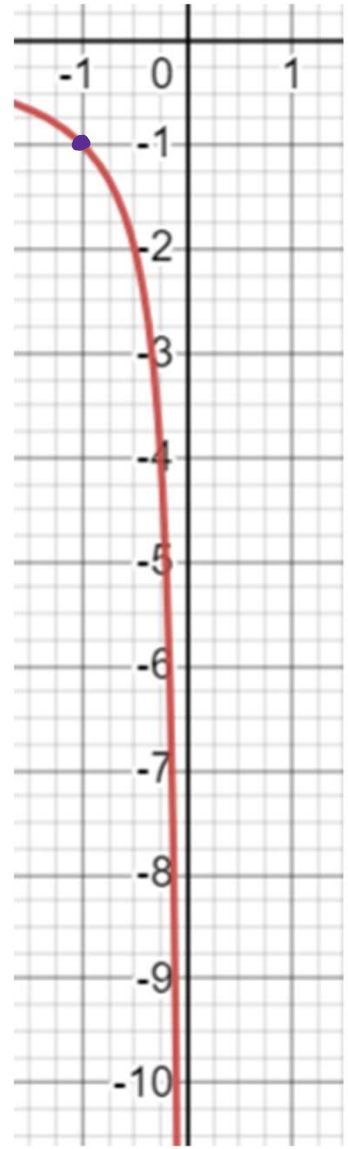
As x gets smaller from the left from negative one to zero, the output gets bigger and bigger in a negative direction.

f(x)=$\frac{1}{x}$ or y= $\frac{1}{x}$ $x \neq 0$

$$\lim_{x \to 0^-} \frac{1}{x} = -\infty$$

$$\frac{-1}{\text{smaller \#}} = \text{bigger \#}$$

$\frac{1}{-1} = -1$

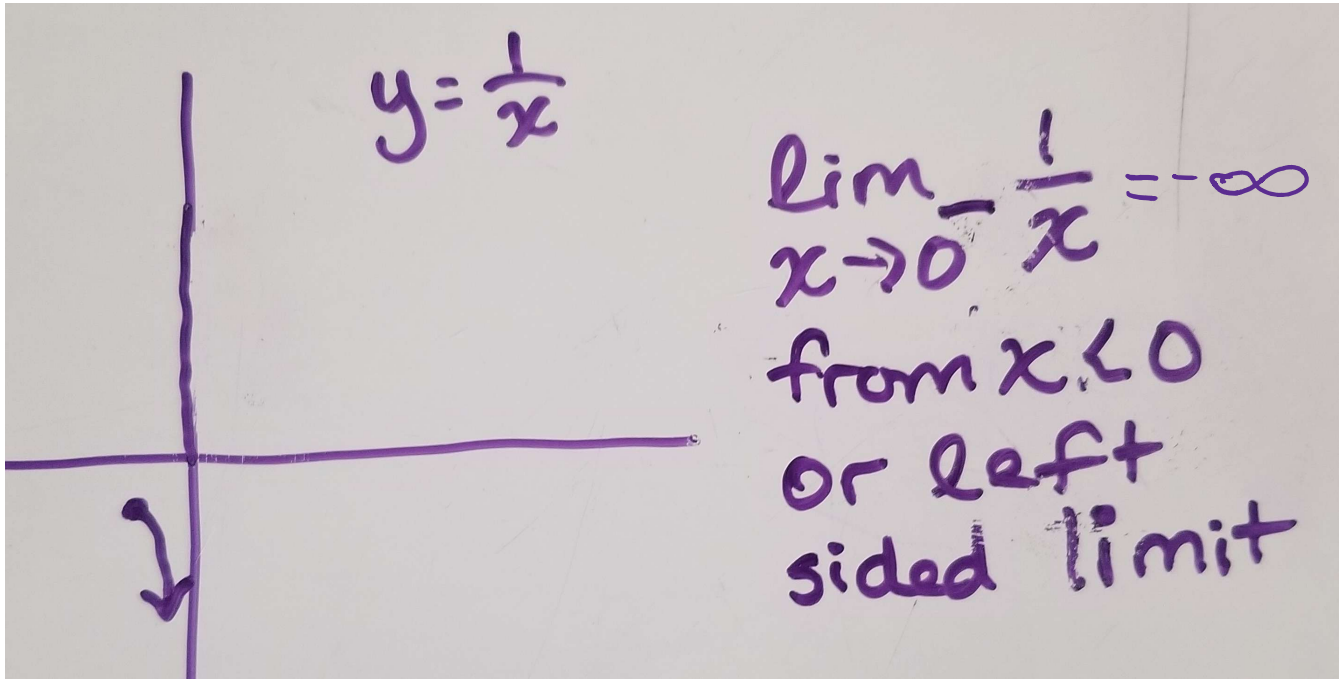


$y = \frac{1}{x}$

$\frac{1}{-\frac{1}{2}} = -2$

$\frac{1}{-\frac{1}{4}} = -4$

$\frac{1}{-\frac{1}{10}} = -10$

$f(x)=\frac{1}{x}$ or $y=\frac{1}{x}$ $x\neq 0$

$$y=\frac{1}{x}$$